

# Mohammad Amin Sadat

Embedded Software Engineer

✉ s.mohammad.amin.sadat@gmail.com    🌐 MohammadAminSadat    in LinkedIn    🌐 mohammadaminsadat.github.io

## Professional Summary

Embedded Software Engineer with 3+ years of experience in safety-critical automotive ADAS systems (ISO 26262, ASIL B/D), holding an M.Sc. in Mechatronics Engineering. Specialises in AEB, ACC, FCW, LDW, and TSR software for BYD, JAC, and GM platforms – covering the complete V-cycle from MiL through MiL/SiL (VTD), HiL (CAN simulation), and vehicle validation with MF4 data recording. Strong background in C/C++, modern C++20 software design, model-based development, AUTOSAR, real-time systems, and modular software architecture. Experienced with CMake, unit testing, static analysis, template-based programming, and Linux development. Additional expertise in ROS-based robotics and applied AI/ML, including MPC+Reinforcement Learning for autonomous path tracking.

## Skills

**Languages:** C (MISRA-aware), C++, Python, Rust, MATLAB/Simulink

**Build & Development:** CMake, GCC, Clang, clang-format, clang-tidy, Git, GDB, Linux, GoogleTest

**Libraries & Frameworks:** Boost, Eigen, Redis, SQLite3, ROS/ROS2, FreeRTOS

**Microcontrollers & Compilers:** ARM Cortex-M, Infineon AURIX TC3xx (Hightec, Tasking), STM32, Embedded Linux, Arduino

**RTOS & Middleware:** FreeRTOS, AUTOSAR (BSW/RTE), ROS/ROS2, Linux Kernel Module

**Communication Protocols:** CAN/CAN-FD, LIN, FlexRay, UDS, USB, UART, SPI, I2C, LoRa, Zigbee, Wi-Fi, BLE, TCP/IP stack

**Testing & Tooling:** GTest, VTD (SiL), CAN-signal HiL simulation, MF4 vehicle data recording, Vector CANoe, Vector CANape, TOUSAN TSmaster, ZLG, Lauterbach, JTAG/SWD, GDB, Git, Jira

**Standards & Methodologies:** ISO 26262 (ASIL B/D), SOTIF awareness, AEB, FCW, LDW, TSR, ACC, Agile/Scrum

**AI / ML:** Deep Learning (PyTorch/TensorFlow), MPC, Reinforcement Learning, edge inference, ROS perception pipelines

## Experience

**Software Motion** - Embedded ADAS Engineer (Decision, Planning & Control Team)

Apr 2025 - Present

- Developed safety-critical embedded software for AEB, FCW, LDW, TSR, and ACC functions on BYD, JAC, and GM vehicles; implemented hierarchical state machines in C with full AUTOSAR RTE integration.
- **Acted as de-facto technical lead for embedded C development:** the team's software lead specialised in model-based design (Simulink); assumed ownership of C coding standards, architecture decisions, and code review for state machine modules across the team - mentored colleagues on hierarchical state machine design, AUTOSAR RTE integration patterns, and embedded coding practices.
- **AEB Module Owner (Zamyad platform):** Owner of the AEB software module across its full lifecycle — managed feature development, triaged and resolved field-reported defects, and maintained a legacy C codebase under active production constraints. Drove incremental improvements to braking threshold logic and target classification heuristics while preserving ISO 26262 ASIL-B and MISRA compliance; coordinated regression runs through MiL/SiL/HiL pipelines after each change to ensure no unintended behaviour was introduced.
- Designed and configured CAN communication stack (DBC signal extraction); validated end-to-end signal flow and timing with Vector CANoe; resolved field bus issues across SiL and vehicle test phases.
- Executed **Module-in-Loop (MiL)** testing with GTest framework to verify individual software unit logic and edge-case behaviour before integration.
- Performed **Software-in-Loop (SiL)** testing using VTD (Virtual Test Drive) - replayed scenario videos in front of the ADAS sensor model and simulated CAN signals to validate algorithm responses in a closed-loop virtual environment.
- Conducted **Hardware-in-Loop (HiL)** validation by injecting simulated CAN signals directly into the physical ECU (AURIX TC379, Tasking compiler), verifying real-time behaviour under representative bus conditions.
- Led **vehicle test campaigns:** recorded ECU output data in MF4 format and fed logs into the company's custom SiL simulator for regression and corner-case analysis; closed feedback loop with firmware team to resolve field issues.

- **From Nov 2025 - Onsite Technical Lead:** managed multi-ECU vehicle test sessions, coordinated data logging strategy, and reduced field-to-fix cycle time through structured issue triage with the firmware team.

#### **Avid Mechanic Hafez Sanaat** - Embedded Mechatronics Engineer

Oct 2021 - Mar 2023

- Developed FreeRTOS firmware in C for a hospital bed controller and a 6-DOF robotic manipulator; implemented motor control, sensor fusion, and fault handling.
- Designed multi-layer PCBs in Altium Designer using STM32 MCUs with I2C/SPI peripheral trees; took boards from schematic to bring-up and production validation.

#### **Teaching Assistant - Embedded & Control Systems**

Shahid Beheshti Univ. & Shiraz Univ.

2018 - 2024

- Led lab sessions on embedded C, Arduino, and real-time control; designed assignments on state machines and PID; mentored 30+ students per semester on hardware debugging.

## Projects

**Trading Engine (C++20)** - Modular trading and backtesting engine developed with modern C++ software engineering practices. Implements market-data handling, technical indicators, CSV parsing, data providers, and the NNX trading strategy. Uses CMake and modular library architecture with GoogleTest, clang-format, clang-tidy, spdlog, fmt, nlohmann/json, SQLite3, Boost, and Eigen.

[GitHub](#)

**Fuzzy Logic Library (Rust)** - Embedded-friendly type-1 fuzzy inference engine implementing Mamdani and Sugeno methods; zero-copy design targeting resource-constrained MCUs. Published open-source.

[fuzzy-logic\\_rs](#)

**MPC+RL Path Tracking for Autonomous Vehicles** - Designed and validated a hybrid Model Predictive Control / deep Reinforcement Learning controller for autonomous lane tracking; achieved stable path following under model uncertainty and dynamic obstacle scenarios in simulation.

MSc Thesis, 2025

**Inverted Pendulum on a Cart** - Real-time embedded control on Arduino using PID and full-state feedback. Tuned controllers via serial debugging; achieved 2 degree steady-state error in hardware validation.

Spring 2023

**IoT Garden - Full Embedded Product** - Custom PCB (FreeCAD), ATmega328P firmware (C++), Wi-Fi telemetry upload, and real-time web dashboard for soil moisture monitoring - complete product lifecycle from schematic to deployment.

Fall 2021

**Pneumatic Sequence Control** - Programmed Siemens S7-1200 PLC in TIA Portal for a 3-cylinder sequencing system using ladder logic and HMI simulation.

Spring 2023

## Education

#### **MS Shahid Beheshti University**, Mechatronics Engineering

Sep 2022 - Feb 2025

- **Rank 1** in Mechatronics Engineering
- Thesis: *Path tracking control of a self-driving car using MPC and Reinforcement Learning* - combined classical and learning-based planners for robust lane-keeping under dynamic constraints

#### **BS Shiraz University**, Mechanical Engineering

Sep 2017 - Feb 2022

- IOT garden project

## Certifications & Awards

- [Motion Planning for Self-Driving Cars \(Coursera\)](#)
- [State Estimation and Localization for Self-Driving Cars \(Coursera\)](#)
- [Generative AI for Everyone \(Coursera\)](#)
- Top 0.5% in Iran's national MSc Mechanical Engineering entrance exam (2023)
- Rank 1 GPA in Mechatronics Engineering at Shahid Beheshti University

## Volunteering

**Core Member of Robotics Club** - Designed and delivered Arduino and MATLAB programming courses; introduced 50+ students to embedded systems and control fundamentals.

Shiraz University,  
Sep 2019 - Sep 2020